

QIUZHEN QUALIFY EXAM FOR ARTIFICIAL INTELLIGENCE

MOCK EXAMINATION NO. 3

Instructions

- This examination consists of four sections, each worth 33 points. Choose exactly three sections and answer all questions in the selected sections.
 - Write your name and student ID on the answer sheet for 1 additional point. The maximum possible score is 100.
 - State the three selected sections at the beginning of the answer sheet. If more than three sections are attempted, only the first three will be graded.
 - Unless stated otherwise, justify every claim and show the essential derivation. Unsupported final formulas receive limited credit.
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A. Machine Learning Theory

1. [8 pts.] Let $\mathcal{H}$ be a finite binary hypothesis class with $|\mathcal{H}| = M$, and let $S \sim P^m$. Under zero-one loss, let $\hat{h} \in \arg \min_{h \in \mathcal{H}} \hat{L}_S(h)$ and $h^* \in \arg \min_{h \in \mathcal{H}} L_P(h)$; no realizability assumption is made. Derive a uniform convergence bound using Hoeffding's inequality and prove a high-probability excess-risk guarantee for $\hat{h}$. Give an explicit sufficient sample size for $L_P(\hat{h}) \leq L_P(h^*) + \varepsilon$ with probability at least $1 - \delta$.

2. [8 pts.] For a fixed sample $S = (x_1, \dots, x_m)$ with $\|x_i\|_2 \leq R$, consider

$$\mathcal{F} = \{x \mapsto w^\top x : \|w\|_2 \leq B\}.$$

Starting from the definition of empirical Rademacher complexity, prove

$$\hat{\mathfrak{R}}_S(\mathcal{F}) \leq \frac{BR}{\sqrt{m}}.$$

State the additional property of a loss needed to transfer this bound to a loss class, and explain why the displayed bound has no explicit ambient-dimension factor.

3. [8 pts.] Let $\ell(w, z)$ be convex in w and ρ -Lipschitz under the Euclidean norm. For a sample $S = (z_1, \dots, z_m)$ define

$$A(S) = \arg \min_w \left\{ \frac{1}{m} \sum_{i=1}^m \ell(w, z_i) + \frac{\lambda}{2} \|w\|_2^2 \right\}, \quad \lambda > 0.$$

If S and S' differ in one example, prove a bound on $\|A(S) - A(S')\|_2$ and hence a uniform replace-one stability bound. State the resulting bound on the expected generalization gap, including all assumptions used.

4. [9 pts.] Let

$$\mathcal{H}_d = \{x \mapsto \mathbf{1}\{w^\top x > 0\} : w \in \mathbb{R}^d\}$$

be the class of homogeneous halfspaces. Prove that $\text{VCdim}(\mathcal{H}_d) = d$. Your upper-bound proof must cover an arbitrary set of $d + 1$ points, not only points in general position.

B. Deep Learning and Reinforcement Learning

1. [8 pts.] Consider a residual block

$$a = W_1 x, \quad h = \phi(a), \quad y = x + W_2 h, \quad J = \mathcal{L}(y),$$

where $x, y \in \mathbb{R}^d$, $W_1 \in \mathbb{R}^{r \times d}$, $W_2 \in \mathbb{R}^{d \times r}$, and ϕ acts coordinatewise. Derive $\nabla_x J$, $\nabla_{W_1} J$, and $\nabla_{W_2} J$ with dimensions. Explain mathematically how the identity path changes gradient propagation. If W_2 is initialized to zero, state which parameter gradients are zero on the first backward pass and which are not.

2. [9 pts.] An affine coupling layer splits $z = (z_a, z_b)$ and defines

$$x_a = z_a, \quad x_b = z_b \odot \exp(s(z_a)) + t(z_a),$$

where s, t have the same dimension as z_b .

- (a) [3 pts.] Derive the inverse transformation.
- (b) [3 pts.] Derive the Jacobian determinant and the exact log-density $\log p_X(x)$ in terms of the base density p_Z .
- (c) [3 pts.] Write the dataset negative-log-likelihood objective and explain why replacing $\exp(s)$ by an unconstrained scale that can be zero invalidates the normalizing-flow likelihood.

3. [10 pts.] For a diffusion process, suppose

$$q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I), \quad x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon,$$

where $\varepsilon \sim \mathcal{N}(0, I)$.

- (a) [3 pts.] Derive $\nabla_{x_t} \log q(x_t | x_0)$ and express it using ε .
 - (b) [4 pts.] Prove that regression to the conditional score has the same minimizer as regression to the marginal score $\nabla_{x_t} \log q_t(x_t)$, up to an additive constant independent of the model.
 - (c) [3 pts.] Derive the equivalent noise-prediction objective and give the conversion from a noise predictor $\varepsilon_\theta(x_t, t)$ to a score predictor.
4. [6 pts.] At a fixed state s , let $A \sim \pi_\theta(\cdot | s)$, $u(A) = \nabla_\theta \log \pi_\theta(A | s)$, and suppose $Q(s, A)$ is known. Consider the policy-gradient estimator $g_b = u(A)(Q(s, A) - b)$ with a scalar state-dependent baseline b . Prove that its expectation is independent of b , and derive the value of b that minimizes $\mathbb{E}[\|g_b\|_2^2 | s]$.

C. Optimization Methods in Artificial Intelligence

1. [11 pts.] Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable and μ -strongly convex. Assume coordinate-wise smoothness:

$$f(x + te_i) \leq f(x) + t \partial_i f(x) + \frac{L_i}{2} t^2, \quad L_i > 0.$$

At each iteration choose coordinate i with probability p_i and set $x^+ = x - (\partial_i f(x)/L_i) e_i$.

- (a) [4 pts.] Derive the conditional expected one-step decrease.
- (b) [4 pts.] For $p_i = L_i / \sum_j L_j$, prove a linear convergence rate in expected function suboptimality.
- (c) [3 pts.] Compare the rate with uniform sampling using $L_{\max}$ and explain when the importance distribution improves the bound.

2. [12 pts.] Let f be convex and L -smooth with minimizer x^* . Consider Nesterov's iteration

$$\begin{aligned} x_{k+1} &= y_k - \frac{1}{L} \nabla f(y_k), & t_{k+1} &= \frac{1 + \sqrt{1 + 4t_k^2}}{2}, \\ y_{k+1} &= x_{k+1} + \frac{t_k - 1}{t_{k+1}} (x_{k+1} - x_k), & t_0 &= 1, \quad y_0 = x_0. \end{aligned}$$

Prove an $O(1/k^2)$ upper bound on $f(x_k) - f(x^*)$ with an explicit constant. The proof must identify a telescoping potential and use the assumptions at the steps where they are needed.

3. [10 pts.] Let f be lower bounded by $f_{\inf}$ and L -smooth, but not necessarily convex. Suppose

$$\mathbb{E}_k[g_k] = \nabla f(x^k), \quad \mathbb{E}_k \left\| g_k - \nabla f(x^k) \right\|^2 \leq \sigma^2, \quad x^{k+1} = x^k - \gamma g_k.$$

For $0 < \gamma \leq 1/L$, derive an upper bound on $K^{-1} \sum_{k=0}^{K-1} \mathbb{E} \left\| \nabla f(x^k) \right\|^2$. Choose a valid K -dependent step size and state the resulting rate and its assumptions.

D. Natural Language Processing

1. [8 pts.] Let $C(w, c)$ be word-context pair counts, $C(w) = \sum_c C(w, c)$, and let negative contexts be sampled k times per positive pair from distribution $q(c)$. Treating each dot product $x_{wc} = u_w^\top v_c$ as an independent scalar parameter, the expected SGNS objective contains

$$C(w, c) \log \sigma(x_{wc}) + kC(w)q(c) \log \sigma(-x_{wc}).$$

Derive the optimal x_{wc} . Show its relation to PMI when $q(c)$ is the empirical context unigram distribution, and state two reasons the matrix of learned finite-dimensional dot products need not equal the full shifted-PMI matrix exactly.

2. [8 pts.] A supervised topic model has K topics, $\theta_d \sim \text{Dir}(\alpha)$, $\phi_k \sim \text{Dir}(\eta)$, token topics $z_{dn} \sim \text{Cat}(\theta_d)$, words $w_{dn} \sim \text{Cat}(\phi_{z_{dn}})$, and

$$p(y_d = 1 \mid \bar{z}_d, \beta) = \sigma(\beta^\top \bar{z}_d), \quad \bar{z}_d = \frac{1}{N_d} \sum_n e_{z_{dn}}.$$

Write the complete joint distribution and derive the gradient of the label negative log-likelihood with respect to β . Explain precisely why the labels can change the learned topics relative to unsupervised LDA.

3. [9 pts.] Let $\varphi : \mathbb{R}^{d_k} \rightarrow \mathbb{R}^r$ be a nonnegative feature map. Define normalized linear attention by

$$Y_i = \frac{\varphi(q_i)^\top \left(\sum_{j=1}^L \varphi(k_j) v_j^\top \right)}{\varphi(q_i)^\top \left(\sum_{j=1}^L \varphi(k_j) \right)}.$$

Give a causal prefix-sum implementation, analyze time and memory complexity in L, r, d_v , and prove that every earlier position can directly influence every later output when the corresponding kernel weight is nonzero. State one limitation relative to exact softmax attention.

4. [8 pts.] An MLN has two unknown atoms $A = \text{AI}(P_1)$ and $B = \text{AI}(P_2)$. The only citation grounding considered is observed true: $\text{Cites}(P_1, P_2)$. It has three weighted formulas:

$$A \Rightarrow B \quad (w_1 = 1.5), \quad A \quad (w_2 = 0.8), \quad B \quad (w_3 = -0.4).$$

The score of a world is the sum of weights of satisfied formula groundings, and $P(x) \propto e^{\text{score}(x)}$. Describe the ground Markov-network structure, enumerate all four world scores, write the partition function, determine the MAP assignment, and compute $P(A = 1, B = 1)/P(A = 0, B = 0)$. Explain the meaning of the negative weight on B under the stated satisfied-grounding convention.